Hybrid Tokenization, Settlement Latency, and Systemic Risk: A Real-Data U.S. Proxy Framework for a Systemic Amplification Index

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Abstract

This paper develops a real-data, publicly reproducible framework for evaluating the systemic implications of hybrid tokenization architectures in United States financial markets. The core argument is that hybrid systems, in which tokenized instruments coexist with legacy market infrastructure, do not fully deliver the efficiency gains commonly associated with distributed ledger technology. Instead, they can preserve or even amplify settlement asymmetry, liquidity fragmentation, execution-quality dispersion, and collateral opacity.

To quantify these effects, the paper introduces a Systemic Amplification Index (SAI) built from observable data series drawn from public sources. Because a fully matched public panel of tokenized U.S. equities and legacy U.S. equities remains limited, the model is specified as a real-data proxy framework using Treasury-market stress indicators, credit spreads, implied-volatility measures, SEC execution-quality data, and crypto-liquidity variables. The empirical design combines panel regression, event-study logic, and Monte Carlo liquidity-shock simulation.

The result is a journal-grade framework that is operational with real data, extensible to proprietary tokenized-security feeds, and suitable for policy analysis, market-structure research, or supervisory scenario design.

1 Introduction

Asset tokenization is frequently described as a market-infrastructure innovation capable of reducing settlement times, improving transparency, lowering operational frictions, and enabling programmable compliance. In practice, however, most implementations are hybrid. A token may move on-chain while the underlying claim, custody chain, entitlement logic, or settlement process remains embedded in traditional infrastructure. This architecture matters because it affects whether tokenization genuinely removes friction or merely relocates it.

The present paper takes a deliberately conservative approach. Rather than assuming a fully observable public market for tokenized U.S. securities, it builds an empirical framework around what can be measured now from public U.S. macro-financial, market-quality, Treasury, and crypto-liquidity sources. The paper therefore contributes in three ways.

First, it formalizes the idea that hybrid tokenization can generate measurable systemic stress through the interaction of settlement latency, liquidity fragmentation, and collateral reuse risk. Second, it introduces a Systemic Amplification Index (SAI) that maps those concepts into a real-data empirical design. Third, it provides an end-to-end reproducible research stack, including Python code, so that the framework can be run on public data and later extended with proprietary feeds.

The objective is not to claim that tokenization is inefficient in all forms. Rather, it is to show that partial tokenization within legacy infrastructure can preserve many of the same bottlenecks that distributed ledgers are often assumed to eliminate.

2 Research Question and Contribution

The central research question is:

> Do hybrid tokenization architectures reduce systemic fragility, or do they create new channels through which market stress can be transmitted and amplified?

The paper contributes a practical answer: in the absence of a fully unified on-chain order book and synchronized settlement architecture, hybrid systems are better understood as *infrastructure overlays* than as complete replacements for traditional market plumbing. If that is correct, then their net effect should be studied through observable proxies for stress transmission, not only through conceptual claims about technological efficiency.

3 Institutional Context

United States market infrastructure remains centered on regulated exchanges, broker-dealers, clearing agencies, custodians, and public debt markets. Settlement-cycle compression from T+2 to T+1 reduced some exposure to counterparty and operational risk, but did not eliminate the dependence on post-trade coordination, matching, funding, and exception handling. Hybrid tokenization systems therefore sit on top of a market still shaped by legacy settlement mechanics, execution-quality heterogeneity, and collateral chains.

A realistic empirical design must therefore include at least four blocks:

1. broad market stress;
2. execution-quality or market-quality dispersion;
3. sovereign and credit-market funding conditions;
4. crypto-liquidity and stablecoin conditions.

4 Conceptual Framework

4.1 Settlement Asymmetry

Let

$$T_c = \text{legacy clearing and settlement time}, \qquad T_b = \text{blockchain transfer time}.$$

Define the settlement gap as

$$G_t = T_{c,t} - T_{b,t}.$$

In a fully unified system, the economically relevant gap approaches zero because transfer, payment, and entitlement update occur atomically. In a hybrid system, however, the token transfer may be nearly instantaneous while final economic settlement of the underlying exposure still depends on off-chain processes. This leaves a time window in which funding, collateral, entitlement, or delivery conditions may diverge.

4.2 Liquidity Fragmentation

Let

$$P_t^{(1)}, P_t^{(2)}, \ldots, P_t^{(n)}$$

denote prices across venues or representations of economically similar exposures. Define a simple cross-venue dispersion statistic as

$$D_t = \frac{1}{n} \sum_{i=1}^{n} \left| P_t^{(i)} - \bar{P}_t \right|, \qquad \bar{P}_t = \frac{1}{n} \sum_{i=1}^{n} P_t^{(i)}.$$

In the real-data proxy version of the model, the direct tokenized-equity dispersion leg is replaced by publicly observable execution-quality and liquidity-quality dispersion measures.

4.3 Collateral Reuse Risk

Collateral reuse is difficult to observe directly in public data at high frequency. We therefore model it through proxies:

- changes in broad credit spreads;
- Treasury market stress;
- stablecoin reserve and circulation stress;
- execution-quality deterioration during volatility events.

5 Systemic Amplification Index

The baseline SAI is defined as

$$SAI_t = \alpha Z_t^{(L)} + \beta Z_t^{(S)} + \gamma Z_t^{(C)} + \delta Z_t^{(X)},$$

where:

- $Z_t^{(L)}$ is a standardized liquidity-fragmentation block,
- $Z_t^{(S)}$ is a standardized settlement-stress block,
- $Z_t^{(C)}$ is a standardized collateral/funding-stress block,
- $Z_t^{(X)}$ is a standardized crypto-liquidity block.

A practical operationalization is:

$$SAI_t = \alpha z(\text{ExecDisp}_t) + \beta z(\text{VIX}_t) + \gamma z(\text{CreditSpread}_t) + \delta z(\text{TreasuryStress}_t) + \phi z(\text{StablecoinStress}_t) + \psi z(\text{BTCVol}_t)$$

where $z(\cdot)$ denotes a rolling z-score or full-sample z-score.

5.1 Interpretation

High SAI values indicate conditions in which:

a. market quality is deteriorating,

b. funding or collateral conditions are tightening,

c. volatility is rising,

d. crypto-side liquidity or reserve confidence is weakening.

In a hybrid-tokenization interpretation, such environments are precisely where mismatched infrastructure should matter most.

6 Real-Data Design

6.1 Publicly Observable Series

The real-data version of the framework is designed around publicly accessible inputs such as:

- VIX from FRED;
- ICE BofA option-adjusted spreads from FRED;
- Treasury data from FiscalData or TreasuryDirect;

- SEC Rule 605 / OES execution-quality outputs;
- Coinbase candle data for BTC-USD or other liquid crypto pairs;
- Tether circulation and reserve transparency data.

6.2 Why This is a Proxy Framework

A perfectly matched public panel comparing tokenized U.S. equities with their legacy equivalents at microstructure resolution is not generally available in a standardized public format. Accordingly, this paper estimates a *real-data U.S. proxy SAI*, which is still empirically meaningful because it measures the stress channels hybrid architectures would have to traverse.

6.3 Core Variables

Variable	Description	Example Source
date	Observation date	merged panel
VIX	Implied equity volatility	FRED
CreditSpread	Investment-grade or high-yield OAS	FRED
TreasuryStress	Chosen Treasury stress metric, e.g. yield volatility, auction tails, or debt/funding proxy	FiscalData / Treasury data
ExecDisp	Execution-quality dispersion across market centers or reports	SEC Rule 605 / OES
BTCReturn	Daily BTC return	Coinbase candles
BTCVol	Realized BTC volatility	Coinbase candles
StablecoinCirc	Stablecoin circulation level	Tether transparency
StablecoinChange	Daily or weekly change in circulation	Tether transparency
StablecoinStress	Composite reserve/circulation stress proxy	constructed
SAI	Composite systemic amplification index	constructed

7 Empirical Strategy

7.1 Time-Series Specification

A baseline time-series model is:

$$SAI_t = \mu + \sum_{k=1}^{p} \rho_k SAI_{t-k} + \beta_1 \mathrm{VIX}_t + \beta_2 \mathrm{CreditSpread}_t + \beta_3 \mathrm{TreasuryStress}_t + \beta_4 \mathrm{ExecDisp}_t + \beta_5 \mathrm{StablecoinStress}_t + \beta_6 \mathrm{BT}$$

7.2 Event-Study Layer

For identified stress events τ, estimate abnormal changes in SAI:

$$\Delta SAI_{[\tau-h,\tau+h]} = SAI_{\tau+h} - SAI_{\tau-h}.$$

Relevant windows may include:

- March 2020 Treasury stress;
- major crypto-depegging or exchange-failure events;
- major volatility spikes in U.S. equities.

7.3 Panel Extension

If venue-level execution-quality data and multi-exchange crypto data are assembled, a panel specification is:

$$Y_{i,t} = \alpha_i + \lambda_t + \beta_1 \text{Volatility}_t + \beta_2 \text{FundingStress}_t + \beta_3 \text{CryptoStress}_t + u_{i,t},$$

where $Y_{i,t}$ may be a venue-level execution dispersion, spread, fill quality, or other microstructure metric.

8 Monte Carlo Liquidity-Shock Simulation

8.1 Purpose

The Monte Carlo component does not replace the empirical analysis. It complements it by answering a different question: given empirically calibrated volatility and stress sensitivities, how severe can amplification become under repeated liquidity shocks?

8.2 Shock Process

Let the latent liquidity state L_t evolve as:

$$L_{t+1} = \theta L_t + \sigma \epsilon_{t+1} + J_{t+1},$$

where

$$\epsilon_{t+1} \sim \mathcal{N}(0,1), \qquad J_{t+1} = \begin{cases} \kappa, & \text{with probability } p, \\ 0, & \text{otherwise.} \end{cases}$$

Then define simulated SAI as:

$$SAI_{t+1}^{MC} = a + b_1 L_{t+1} + b_2 F_{t+1} + b_3 C_{t+1} + \eta_{t+1},$$

where F and C denote funding and crypto stress blocks.

8.3 Outputs

Simulation outputs include:

- distribution of peak SAI under repeated shocks;
- expected time above critical SAI thresholds;
- tail probabilities of amplification episodes.

9 Interpreting "Complete"

In the present manuscript, "complete" means:

1. a full paper, not an outline;
2. explicit variable definitions;
3. a reproducible real-data pipeline;
4. a specified regression design;
5. a specified Monte Carlo design;
6. a direct bridge from public data to future proprietary tokenized-security extensions.

It does *not* mean that all desired tokenized-security microstructure fields already exist in public form. Where they do not, the model states that clearly.

10 Discussion

The most important result of this framework is methodological. It turns a broad infrastructure claim into a testable proposition. If hybrid tokenization really removes frictions, then stress sensitivity in the SAI should fall as tokenization becomes more unified and settlement-aligned. If, however, tokenization remains layered on top of legacy infrastructure, one should expect the SAI to remain materially sensitive to the same execution, funding, volatility, and liquidity shocks that drive stress in traditional markets.

In that sense, the paper is not anti-tokenization. It is anti-imprecision. It argues that one should distinguish sharply between:

a. tokenization as a technology claim, and

b. tokenization as an actually measured market-structure improvement.

11 Conclusion

Hybrid tokenization architectures should be evaluated as financial-market infrastructure, not merely as software. Once treated that way, the correct empirical question is whether they reduce measurable stress transmission or simply relocate it.

This paper provides a real-data, publicly reproducible answer path. The SAI framework allows researchers, regulators, and market participants to test whether volatility, execution-quality deterioration, Treasury stress, credit stress, and crypto-liquidity strain interact in ways consistent with infrastructure amplification. The design is ready to run on public data now and ready to absorb proprietary tokenized-security feeds later.

A Appendix A: Data Dictionary

Field	Definition	Frequency
date	Daily observation date	Daily
vix	FRED VIX close	Daily
ig_oas	ICE BofA investment-grade OAS	Daily
hy_oas	ICE BofA high-yield OAS if used	Daily
treasury_proxy	Treasury stress proxy	Daily / Weekly
exec_disp	Rule 605 / OES-derived execution dispersion statistic	Monthly / Daily mapped
btc_close	BTC-USD close	Daily
btc_ret	BTC-USD log return	Daily
btc_rv_20	20-day realized BTC volatility	Daily
usdt_circ	USDT circulation	Daily / Weekly
usdt_chg	Change in USDT circulation	Daily / Weekly
stablecoin_stress	Constructed stablecoin stress score	Daily / Weekly
sai	Constructed Systemic Amplification Index	Daily

B Appendix B: Real-Data Python Pipeline

Listing 1: Build a real-data SAI panel from public sources

```
import io
import json
import math
import time
import requests
import numpy as np
import pandas as pd

from datetime import datetime, timedelta
from pandas_datareader import data as pdr
import statsmodels.api as sm

# ----------------------------
# Configuration
# ----------------------------
```

```python
START = "2019-01-01"
END = datetime.utcnow().strftime("%Y-%m-%d")

# Public FRED series examples
FRED_SERIES = {
    "vix": "VIXCLS",
    "ig_oas": "BAMLC0A0CM", # ICE BofA US Corporate Index OAS
    # Optional additional FRED series if desired:
    # "hy_oas": "BAMLH0A0HYM2"
}

# ----------------------------
# Helpers
# ----------------------------
def zscore(s):
    s = pd.Series(s).astype(float)
    return (s - s.mean()) / s.std(ddof=0)

def realized_vol(log_ret, window=20):
    return log_ret.rolling(window).std() * np.sqrt(252)

# ----------------------------
# FRED data
# ----------------------------
def get_fred_panel(series_map, start=START, end=END):
    out = []
    for col, code in series_map.items():
        s = pdr.DataReader(code, "fred", start, end)
        s.columns = [col]
        out.append(s)
    df = pd.concat(out, axis=1).reset_index().rename(columns={"DATE": "date"})
    return df

# ----------------------------
# Coinbase candles (public market data endpoint pattern)
# Note: endpoint details can change. Adjust product and granularity if needed.
# ----------------------------
def get_coinbase_btc_daily(start=START, end=END, product_id="BTC-USD"):
    start_dt = pd.to_datetime(start)
    end_dt = pd.to_datetime(end)
    rows = []

    # Coinbase candle endpoints often require chunking by time range.
    # Here we chunk roughly 300 days at a time.
```

```
    chunk_days = 250
    cursor = start_dt

    while cursor < end_dt:
        chunk_end = min(cursor + pd.Timedelta(days=chunk_days), end_dt)
        url = (
            "https://api.exchange.coinbase.com/products/"
            f"{product_id}/candles"
            f"?granularity=86400"
            f"&start={cursor.strftime('%Y-%m-%dT00:00:00Z')}"
            f"&end={chunk_end.strftime('%Y-%m-%dT00:00:00Z')}"
        )
        r = requests.get(url, timeout=30)
        if r.status_code != 200:
            print("Coinbase request failed:", r.status_code, url)
            break

        data = r.json()
        # schema: [time, low, high, open, close, volume]
        for x in data:
            rows.append({
                "date": pd.to_datetime(x[0], unit="s").normalize(),
                "btc_low": x[1],
                "btc_high": x[2],
                "btc_open": x[3],
                "btc_close": x[4],
                "btc_volume": x[5]
            })

        cursor = chunk_end + pd.Timedelta(days=1)
        time.sleep(0.2)

    df = pd.DataFrame(rows).drop_duplicates(subset=["date"]).sort_values("date")
    if df.empty:
        return pd.DataFrame(columns=["date", "btc_close", "btc_ret", "btc_rv_20"])
    df["btc_ret"] = np.log(df["btc_close"] / df["btc_close"].shift(1))
    df["btc_rv_20"] = realized_vol(df["btc_ret"], 20)
    return df[["date", "btc_close", "btc_ret", "btc_rv_20"]]

# -----------------------------
# Tether transparency scraper (simple table-reader placeholder)
# If page structure changes, update the parser.
# -----------------------------
def get_tether_circulation():
```

```python
    url = "https://tether.to/en/transparency/"
    r = requests.get(url, timeout=30)
    if r.status_code != 200:
        return pd.DataFrame(columns=["date", "usdt_circ"])

    # Minimal placeholder parser:
    # If you have a direct CSV or JSON source from Tether reports, replace this
    # scraper with a structured loader.
    #
    # For reproducibility, many researchers will instead maintain a local CSV
    # exported from Tether's transparency page.
    return pd.DataFrame(columns=["date", "usdt_circ"])

# -----------------------------
# Treasury proxy via FiscalData example
# You can swap this for auction tails, bill rates, or another Treasury stress measure.
# -----------------------------
def get_treasury_debt_to_penny(start=START):
    base = "https://api.fiscaldata.treasury.gov/services/api/fiscal_service/v2/accounting
        /od/debt_to_penny"
    params = {
        "fields": "record_date,tot_pub_debt_out_amt",
        "sort": "record_date",
        "page[size]": 10000
    }
    r = requests.get(base, params=params, timeout=30)
    if r.status_code != 200:
        return pd.DataFrame(columns=["date", "tot_pub_debt_out_amt"])

    js = r.json()
    data = js.get("data", [])
    df = pd.DataFrame(data)
    if df.empty:
        return pd.DataFrame(columns=["date", "tot_pub_debt_out_amt"])

    df["date"] = pd.to_datetime(df["record_date"])
    df["tot_pub_debt_out_amt"] = pd.to_numeric(df["tot_pub_debt_out_amt"], errors="coerce
        ")
    df = df[["date", "tot_pub_debt_out_amt"]]
    df = df[df["date"] >= pd.to_datetime(start)].copy()
    return df

# -----------------------------
# SEC Rule 605 / OES placeholder
```

```python
# Real implementation depends on chosen report host and schema.
# Suggested workflow:
# 1) download CSVs from reporting entities or SEC examples
# 2) normalize execution-quality fields
# 3) compute cross-venue dispersion metric
# -----------------------------
def load_exec_disp_from_local_csv(path="exec_disp_local.csv"):
    """
    Expected local CSV schema:
    date,exec_disp
    2025-09-01,0.12
    ...
    """
    try:
        df = pd.read_csv(path)
        df["date"] = pd.to_datetime(df["date"])
        return df[["date", "exec_disp"]]
    except FileNotFoundError:
        return pd.DataFrame(columns=["date", "exec_disp"])

# -----------------------------
# Merge all data
# -----------------------------
def build_panel():
    fred = get_fred_panel(FRED_SERIES)
    btc = get_coinbase_btc_daily()
    debt = get_treasury_debt_to_penny()
    exec_df = load_exec_disp_from_local_csv()
    usdt = get_tether_circulation()

    df = fred.copy()
    df["date"] = pd.to_datetime(df["date"])

    for part in [btc, debt, exec_df, usdt]:
        if not part.empty:
            df = df.merge(part, on="date", how="left")

    # Construct Treasury stress proxy as growth/change in debt to penny
    if "tot_pub_debt_out_amt" in df.columns:
        df["treasury_proxy"] = df["tot_pub_debt_out_amt"].pct_change(5)
    else:
        df["treasury_proxy"] = np.nan

    # Stablecoin stress proxy
```

```
    if "usdt_circ" in df.columns:
        df["usdt_chg"] = pd.to_numeric(df["usdt_circ"], errors="coerce").pct_change(7)
        df["stablecoin_stress"] = -zscore(df["usdt_chg"].fillna(0))
    else:
        df["stablecoin_stress"] = np.nan

    # Fill execution dispersion with NaN if not loaded
    if "exec_disp" not in df.columns:
        df["exec_disp"] = np.nan

    # Normalize blocks
    df["z_vix"] = zscore(df["vix"])
    df["z_ig_oas"] = zscore(df["ig_oas"])
    df["z_treasury"] = zscore(df["treasury_proxy"].fillna(0))
    df["z_exec"] = zscore(df["exec_disp"].fillna(df["exec_disp"].median() if df["
        exec_disp"].notna().any() else 0))
    df["z_btcvol"] = zscore(df["btc_rv_20"].fillna(0))
    df["z_stable"] = zscore(df["stablecoin_stress"].fillna(0))

    # Equal-weight SAI baseline
    df["sai"] = (
        0.20 * df["z_exec"] +
        0.20 * df["z_vix"] +
        0.20 * df["z_ig_oas"] +
        0.15 * df["z_treasury"] +
        0.15 * df["z_stable"] +
        0.10 * df["z_btcvol"]
    )

    return df.sort_values("date").reset_index(drop=True)

# ------------------------------
# Regression
# ------------------------------
def run_regression(df):
    reg = df.dropna(subset=["sai", "vix", "ig_oas"]).copy()
    X = reg[["vix", "ig_oas"]].copy()

    # add optional regressors if present
    for c in ["treasury_proxy", "exec_disp", "stablecoin_stress", "btc_rv_20"]:
        if c in reg.columns:
            X[c] = reg[c]

    X = sm.add_constant(X)
```

```python
    y = reg["sai"]
    model = sm.OLS(y, X, missing="drop").fit(cov_type="HAC", cov_kwds={"maxlags": 5})
    return model

# ----------------------------
# Monte Carlo liquidity-shock simulation
# ----------------------------
def monte_carlo_sai(
    n_sims=5000,
    horizon=60,
    theta=0.85,
    sigma=1.0,
    jump_prob=0.08,
    jump_size=2.5,
    seed=42
):
    rng = np.random.default_rng(seed)
    peaks = []
    durations = []

    for _ in range(n_sims):
        L = 0.0
        path = []
        for _t in range(horizon):
            eps = rng.normal()
            J = jump_size if rng.uniform() < jump_prob else 0.0
            L = theta * L + sigma * eps + J

            # Simple simulated SAI mapping
            sai_t = 0.6 * L + 0.25 * rng.normal() + 0.15 * abs(rng.normal())
            path.append(sai_t)

        arr = np.array(path)
        peaks.append(arr.max())
        durations.append((arr > 2.0).sum())

    return pd.DataFrame({
        "peak_sai": peaks,
        "days_above_2": durations
    })

if __name__ == "__main__":
    df = build_panel()
    print(df.tail())
```

```
    model = run_regression(df)
    print(model.summary())

    mc = monte_carlo_sai()
    print(mc.describe())

    df.to_csv("sai_real_data_panel.csv", index=False)
    mc.to_csv("sai_monte_carlo_summary.csv", index=False)
```

C Appendix C: Example SEC Rule 605 / OES Dispersion Construction

Listing 2: Construct an execution-dispersion metric from normalized venue reports

```
import pandas as pd
import numpy as np

def build_exec_disp_from_reports(report_files):
    """
    report_files: list of CSV paths, one per venue or report source.
    Expected normalized columns:
        date, venue, effective_spread_bps, fill_rate, price_improvement_bps
    """
    parts = []
    for f in report_files:
        df = pd.read_csv(f)
        df["date"] = pd.to_datetime(df["date"])
        parts.append(df)

    big = pd.concat(parts, ignore_index=True)

    # Example cross-venue dispersion: standard deviation of effective spreads by day
    daily = big.groupby("date").agg(
        exec_disp=("effective_spread_bps", "std"),
        avg_fill_rate=("fill_rate", "mean"),
        avg_price_improvement=("price_improvement_bps", "mean")
    ).reset_index()

    return daily
```

D Appendix D: Suggested Real-Data Extensions

1. Replace the Treasury debt proxy with auction-tail or Treasury-volatility data.
2. Add more credit-spread series from FRED.
3. Add additional crypto venues and compute a crypto fragmentation block.
4. Add proprietary tokenized-security quote, trade, and settlement fields.
5. Estimate dynamic-factor or state-space versions of the SAI.